

A non-atomic polydisc route in the Herglotz–Caratheodory problem

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Source Question

Bhattacharyya–Bhowmik–Kumar ask whether Herglotz’s representation and Caratheodory’s approximation are equivalent on the polydisc or on the symmetrized polydisc [1, p. 19].

measure η on $\mathbb{G}_d$ satisfying

$$\int_{b\mathbb{G}_d} g d\eta = 0 \text{ for all } g \in C(b\mathbb{G}_d) \cap \left(H^2(\mathbb{G}_d) + \overline{H^2(\mathbb{G}_d)} \right)^\perp$$

such that

$$f(\mathbf{s}) = \int_{b\mathbb{G}_d} [2S_{\mathbb{G}_d}(\mathbf{s}, \mathbf{t}) - 1] d\eta(\mathbf{t})$$

for every $\mathbf{s}$ from $\mathbb{G}_d$.

Carathéodory’s approximation theorem for the symmetrized polydisc has been found in [12]. Whether Herglotz’s representation and Carathéodory’s approximation are equivalent on the polydisc or on the symmetrized polydisc remains an open problem.

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This packet does not solve the full two-domain question. It proves a sharp obstruction to the most direct disc-style route from the Koranyi–Pukanszky Herglotz representation on the polydisc to Caratheodory approximation. In fact, the obstruction is stronger than finite atomicity: every positive Koranyi–Pukanszky determining measure has Haar marginals along all mixed-sign characters, and hence is nonatomic.

A further push identifies the exact non-atomic replacement for this failed atomic mechanism. On the polydisc, locally uniform approximation by Cayley rational inner Herglotz functions is equivalent to weak-star density of the corresponding rational-inner Clark measures in the Koranyi–Pukanszky determining-measure simplex.

The final section gives a two-way polydisc route: Caratheodory approximation implies the normalized Koranyi–Pukanszky representation by the same scaling/Helly mechanism used in the source paper, while the Koranyi–Pukanszky representation plus the standard Pfister–Rudin–Stout

finite rationalization step yields Caratheodory approximation without using atomic determining measures. A separate stress test explains why the same argument does not yet descend to the symmetrized polydisc.

Definitions

For $d \geq 2$, let m_d denote Haar probability measure on $\mathbb{T}^d$. In the Koranyi–Pukanszky representation for the polydisc, the representing positive measure μ on $\mathbb{T}^d$ must satisfy the determining moment condition

$$\int_{\mathbb{T}^d} u_1^{n_1} \cdots u_d^{n_d} d\mu(u) = 0$$

whenever the multi-index $n = (n_1, \dots, n_d) \in \mathbb{Z}^d$ has mixed signs, that is, whenever n is neither coordinatewise nonnegative nor coordinatewise nonpositive. Write $\text{RP}(\mathbb{T}^d)$ for the positive finite measures satisfying this condition.

Proof Intuition

The one-variable proof has point masses available: finite Blaschke products correspond, after the inverse Cayley transform, to finite atomic Herglotz measures. The polydisc moment condition destroys that mechanism. If $\mu \in \text{RP}(\mathbb{T}^d)$, then every mixed-sign character $u \mapsto u^n$ has all nonzero moments equal to zero under μ . Thus its pushforward is Haar measure on $\mathbb{T}$. This forces μ to give zero mass to every level set of every mixed-sign character, so in particular μ has no atoms.

Theorem 1 (Haar marginals). *Let $d \geq 2$, and let $\mu \in \text{RP}(\mathbb{T}^d)$. If $n = (n_1, \dots, n_d) \in \mathbb{Z}^d$ has at least one positive coordinate and at least one negative coordinate, define*

$$\chi_n : \mathbb{T}^d \rightarrow \mathbb{T}, \quad \chi_n(u) = u_1^{n_1} \cdots u_d^{n_d}.$$

Then

$$(\chi_n)_\# \mu = \mu(\mathbb{T}^d) m_1,$$

where m_1 is Haar probability measure on $\mathbb{T}$.

Proof. Let $\nu = (\chi_n)_\# \mu$. For every $k \in \mathbb{Z} \setminus \{0\}$, the multi-index kn also has at least one positive and one negative coordinate. Therefore the Koranyi–Pukanszky determining condition gives

$$\int_{\mathbb{T}} \zeta^k d\nu(\zeta) = \int_{\mathbb{T}^d} \chi_n(u)^k d\mu(u) = \int_{\mathbb{T}^d} u^{kn} d\mu(u) = 0.$$

Hence all nonzero Fourier coefficients of ν vanish. Its zeroth Fourier coefficient is $\nu(\mathbb{T}) = \mu(\mathbb{T}^d)$. By uniqueness of Fourier coefficients for finite Borel measures on $\mathbb{T}$,

$$\nu = \mu(\mathbb{T}^d) m_1.$$

□

Corollary 1 (No mass on mixed-character level sets). *Let $d \geq 2$, let $\mu \in \text{RP}(\mathbb{T}^d)$, and let $n \in \mathbb{Z}^d$ have mixed signs. Then*

$$\mu\{u \in \mathbb{T}^d : \chi_n(u) = \lambda\} = 0$$

for every $\lambda \in \mathbb{T}$.

Proof. By the theorem, the pushforward $(\chi_n)_\# \mu$ is a scalar multiple of nonatomic Haar measure. Therefore each singleton $\{\lambda\} \subset \mathbb{T}$ has pushforward mass 0, which is exactly the displayed measure. $\square$

Corollary 2 (Diffuse determining measures). *For $d \geq 2$, every positive finite measure in $\text{RP}(\mathbb{T}^d)$ is nonatomic. In particular, $\text{RP}(\mathbb{T}^d)$ contains no nonzero positive measure with countable support, and no finite atomic probability measures.*

Proof. Fix distinct indices $i \neq j$. The character $\chi(u) = u_i \overline{u_j}$ is mixed-sign. If $a \in \mathbb{T}^d$, then $\{a\} \subseteq \chi^{-1}\{\chi(a)\}$, and the previous corollary gives $\mu(\{a\}) = 0$. Thus μ has no atoms. The countable-support and finite-atomic conclusions follow immediately. $\square$

The Clark-Measure Replacement

Let

$$\mathfrak{H}_0(\mathbb{D}^d) = \{F \in \text{Hol}(\mathbb{D}^d) : \text{Re } F \geq 0, F(0) = 1\}.$$

Let $\mathcal{R}_0(\mathbb{D}^d)$ be the set of normalized Cayley rational inner Herglotz functions

$$F = \frac{1 + \phi}{1 - \phi},$$

where ϕ is rational inner on $\mathbb{D}^d$ and $\phi(0) = 0$. For $F \in \mathfrak{H}_0(\mathbb{D}^d)$, let $\mu_F \in \text{RP}(\mathbb{T}^d)$ denote its unique Koranyi–Pukanszky representing probability measure. If $F \in \mathcal{R}_0(\mathbb{D}^d)$, this measure is the Aleksandrov–Clark measure of the rational inner function ϕ at the parameter 1.

Theorem 2 (Clark density reformulation). *For $d \geq 2$, the following are equivalent.*

1. *The set $\mathcal{R}_0(\mathbb{D}^d)$ is dense in $\mathfrak{H}_0(\mathbb{D}^d)$ for locally uniform convergence.*
2. *The set*

$$\{\mu_F : F \in \mathcal{R}_0(\mathbb{D}^d)\}$$

is weak-star dense in the probability measures in $\text{RP}(\mathbb{T}^d)$.

Proof. Assume first that $\mathcal{R}_0(\mathbb{D}^d)$ is locally uniformly dense in $\mathfrak{H}_0(\mathbb{D}^d)$. Let $\mu \in \text{RP}(\mathbb{T}^d)$ be a probability measure, and let

$$F(z) = \int_{\mathbb{T}^d} (2S_{\mathbb{D}^d}(z, u) - 1) d\mu(u).$$

Choose $F_n \in \mathcal{R}_0(\mathbb{D}^d)$ with $F_n \rightarrow F$ locally uniformly, and write $\mu_n = \mu_{F_n}$. The probability measures on the compact metric space $\mathbb{T}^d$ are weak-star compact. If $\mu_{n_j} \rightarrow \nu$ weak-star along a subsequence, then for each fixed $z \in \mathbb{D}^d$, continuity of $u \mapsto 2S_{\mathbb{D}^d}(z, u) - 1$ gives

$$F(z) = \lim_j F_{n_j}(z) = \int_{\mathbb{T}^d} (2S_{\mathbb{D}^d}(z, u) - 1) d\nu(u).$$

The class $\text{RP}(\mathbb{T}^d)$ is weak-star closed, so $\nu \in \text{RP}(\mathbb{T}^d)$. Uniqueness in the Koranyi–Pukanszky representation gives $\nu = \mu$. Thus every weak-star cluster point of (μ_n) is μ , and hence $\mu_n \rightarrow \mu$.

Conversely, assume weak-star density of the measures μ_F . For $F \in \mathfrak{H}_0(\mathbb{D}^d)$, choose $F_n \in \mathcal{R}_0(\mathbb{D}^d)$ with $\mu_{F_n} \rightarrow \mu_F$ weak-star. Since the kernel $(z, u) \mapsto 2S_{\mathbb{D}^d}(z, u) - 1$ is continuous in u and uniformly bounded for z in a fixed compact subset of $\mathbb{D}^d$, the integral representations give $F_n \rightarrow F$ uniformly on compact subsets of $\mathbb{D}^d$. $\square$

Corollary 3 (Known positive zones). *The Clark-density condition above holds in every setting where Caratheodory approximation by rational inner functions is already known. In particular it holds for the Schur class on the polydisc by the Pfister–Rudin–Stout approximation theorem, as recalled in [6, Sec. 5] and originally proved in [8], and with finite Taylor-section matching for the Agler class by Bhowmik–Putinar [6]. Since the Agler and Schur classes agree on the bidisc by Ando’s theorem, this gives a complete positive bidisc approximation statement in the usual Schur, equivalently Herglotz via Cayley transform, sense.*

Proof. If $\theta = (F-1)/(F+1)$ is the Cayley transform of $F \in \mathfrak{H}_0(\mathbb{D}^d)$, rational-inner approximation of θ by functions vanishing at the origin gives approximation of F by members of $\mathcal{R}_0(\mathbb{D}^d)$ after applying the inverse Cayley transform. The theorem then gives weak-star density of the corresponding Clark measures. The cited approximation theorems provide the stated classes. $\square$

A Two-Way Polydisc Route

We isolate the two assertions involved in the polydisc case. Let KP_d denote the normalized Koranyi–Pukanszky representation: every $F \in \mathfrak{H}_0(\mathbb{D}^d)$ has a representing probability measure $\mu_F \in \text{RP}(\mathbb{T}^d)$. Let CA_d^0 denote the zero-preserving Caratheodory approximation statement: every Schur function $\theta : \mathbb{D}^d \rightarrow \overline{\mathbb{D}}$ with $\theta(0) = 0$ is a locally uniform limit of rational inner functions θ_n with $\theta_n(0) = 0$. This normalized form is equivalent to the usual Caratheodory approximation statement by a disk automorphism.

We shall use the following standard finite rationalization lemma of Pfister–Rudin–Stout. If g is holomorphic on a neighbourhood of $\overline{\mathbb{D}^d}$, $g(0) = 0$, and $\|g\|_{\overline{\mathbb{D}^d}} \leq 1$, then g is locally uniformly approximable by rational inner functions vanishing at the origin. Indeed, after replacing g by a small dilation and a polynomial p with $p(0) = 0$ and $\|p\|_{\overline{\mathbb{D}^d}} < 1$, one uses the reflected-polynomial formula

$$\Phi_N(z) = \frac{p(z) + z_1 \cdots z_d z^N}{1 + z_1 \cdots z_d p^{*N}(z)},$$

where $N = (N, \dots, N)$ is large and

$$p^{*N}(z) = z^N \overline{p(1/\bar{z})}.$$

The denominator is nonzero on $\mathbb{D}^d$, $\Phi_N(0) = 0$, $|\Phi_N| = 1$ on $\mathbb{T}^d$, and $\Phi_N \rightarrow p$ locally uniformly as $\min_j N_j \rightarrow \infty$.

Theorem 3 (Polydisc implications). *For every $d \geq 1$, CA_d^0 implies KP_d . Conversely, KP_d , together with the Pfister–Rudin–Stout finite rationalization lemma above, implies CA_d^0 .*

Proof. First assume CA_d^0 , and let $F \in \mathfrak{H}_0(\mathbb{D}^d)$. Set

$$\theta = \frac{F-1}{F+1}.$$

Then θ is Schur and $\theta(0) = 0$. Choose rational inner θ_n with $\theta_n(0) = 0$ and $\theta_n \rightarrow \theta$ locally uniformly. Pick $0 < r_n < 1$ with $r_n \rightarrow 1$, slowly enough that

$$G_n = \frac{1 + r_n \theta_n}{1 - r_n \theta_n}$$

converges to F locally uniformly. Each G_n is holomorphic on a neighbourhood of $\overline{\mathbb{D}^d}$, has positive real part, and satisfies $G_n(0) = 1$. Hence $h_n = \text{Re } G_n|_{\mathbb{T}^d}$ is a continuous positive function with

$\int h_n dm_d = 1$. Because G_n has only nonnegative Fourier frequencies and $\overline{G_n}$ has only nonpositive Fourier frequencies, the measure $h_n dm_d$ annihilates all mixed-sign characters; that is, $h_n dm_d \in \text{RP}(\mathbb{T}^d)$. The Cauchy–Szego formula gives

$$G_n(z) = \int_{\mathbb{T}^d} (2S_{\mathbb{D}^d}(z, u) - 1) h_n(u) dm_d(u).$$

By weak-star compactness of probability measures on $\mathbb{T}^d$, a subnet of $h_n dm_d$ converges to a probability measure μ . The class $\text{RP}(\mathbb{T}^d)$ is weak-star closed, so $\mu \in \text{RP}(\mathbb{T}^d)$, and passing to the limit in the displayed integral gives the desired representation of F . This proves KP_d .

Conversely assume KP_d , and let θ be Schur with $\theta(0) = 0$. Put $F = (1 + \theta)/(1 - \theta)$, and let $\mu \in \text{RP}(\mathbb{T}^d)$ be its representing probability measure. Let K_N be the product Fejer kernel on $\mathbb{T}^d$, and set $\mu_N = K_N * \mu$. Then μ_N is a positive absolutely continuous probability measure, $\mu_N \in \text{RP}(\mathbb{T}^d)$, and $\mu_N \rightarrow \mu$ weak-star. Write $d\mu_N = h_N dm_d$, and define

$$F_N(z) = \int_{\mathbb{T}^d} (2S_{\mathbb{D}^d}(z, u) - 1) h_N(u) dm_d(u).$$

Since h_N is a trigonometric polynomial, F_N is a holomorphic polynomial. The determining condition gives $\text{Re } F_N \geq 0$, $F_N(0) = 1$, and $F_N \rightarrow F$ locally uniformly. Therefore

$$\theta_N = \frac{F_N - 1}{F_N + 1}$$

is Schur, holomorphic on a neighbourhood of $\overline{\mathbb{D}^d}$, satisfies $\theta_N(0) = 0$, and converges to θ locally uniformly. Applying the Pfister–Rudin–Stout finite rationalization lemma to each θ_N and diagonalizing gives rational inner functions vanishing at the origin that converge locally uniformly to θ . This proves CA_d^0 . $\square$

Corollary 4 (Polydisc status). *For the polydisc, the normalized Koranyi–Pukanszky representation and zero-preserving Caratheodory approximation are equivalent over the standard Pfister–Rudin–Stout finite rationalization lemma. The proof of the Herglotz-to-Caratheodory direction is necessarily non-atomic for $d \geq 2$: Fejer smoothing inside $\text{RP}(\mathbb{T}^d)$ replaces the finite point-mass approximation used in the disc.*

Symmetrized-Polydisc Stress Test

The preceding proof isolates why the polydisc is tractable: $\mathbb{T}^d$ is a compact abelian group, and product Fejer convolution turns an arbitrary positive $\text{RP}(\mathbb{T}^d)$ -measure into positive continuous densities while preserving the mixed-sign annihilation conditions.

For the symmetrized polydisc $\mathbb{G}_d = s(\mathbb{D}^d)$, the source paper uses the Szego kernel

$$S_{\mathbb{G}_d}(s(z), s(w)) = \prod_{i,j=1}^d (1 - z_i \overline{w_j})^{-1}$$

and the boundary measure $\nu = s_{\#}(|J|^2 dm_d)$ on $b\mathbb{G}_d = s(\mathbb{T}^d)$ [1, p. 19]. Bhowmik–Kumar prove Caratheodory approximation by rational inner functions on quotient domains $\theta(\mathbb{D}^d)$, including the symmetrized polydisc [3, Theorem C]. Their proof pulls functions back to $\mathbb{D}^d$, averages to a G -invariant polynomial, and then applies a Rudin–Stout/Pfister reflected-polynomial construction before descending through the quotient map.

This gives a useful one-sided check. If zero-preserving rational-inner Caratheodory approximation is available on $\mathbb{G}_d$, then the normalized Szego-kernel Herglotz representation follows by the same scaling argument as in the first half of the polydisc theorem: approximate the Cayley transform $(F - 1)/(F + 1)$, replace each approximant θ_n by $r_n\theta_n$, set $G_n = (1 + r_n\theta_n)/(1 - r_n\theta_n)$, and take a weak-star limit of the measures $\operatorname{Re} G_n d\nu$. Since $\operatorname{Re} G_n \in H^2(\mathbb{G}_d) + \overline{H^2(\mathbb{G}_d)}$, these measures satisfy the source paper's determining condition.

The reverse implication is where the polydisc proof does not descend. A representing measure $\eta \in \operatorname{RP}(b\mathbb{G}_d)$ need not come with an intrinsic convolution calculus on $b\mathbb{G}_d$. The quotient boundary is not a compact abelian group; in fact Bhowmik–Kumar point out, using Morton's theorem, that the symmetrized-polydisc boundary is non-orientable for even d , so Rudin's compact-group convexity machinery cannot be used there [3, p. 10]. Pulling η back to an S_d -invariant measure on $\mathbb{T}^d$ also does not immediately solve the problem: Fejer smoothing of such a lift must still descend to positive measures annihilating

$$C(b\mathbb{G}_d) \cap (H^2(\mathbb{G}_d) + \overline{H^2(\mathbb{G}_d)})^\perp,$$

and must interact correctly with the weighted Szego measure $s_\#(|J|^2 dm_d)$, not only with the quotient Haar pushforward $s_\#m_d$.

This obstruction is not merely cosmetic: the naive pullback-then-Fejer procedure fails already for $\mathbb{G}_2$. Write $s = z_1 + z_2$, $p = z_1 z_2$, and let

$$q(s, p) = \frac{s^2}{p} - 1.$$

On the distinguished boundary,

$$q \circ s = 1 + z_1 \bar{z}_2 + z_2 \bar{z}_1.$$

The sign-representation Hardy space of $\mathbb{G}_2$ is modelled on the symmetric pullbacks with weighted boundary measure $|z_1 - z_2|^2 dm_2$. Put $a = z_1 \bar{z}_2$. Then

$$|z_1 - z_2|^2 = 2 - a - a^{-1}$$

and

$$(q \circ s)|z_1 - z_2|^2 = (1 + a + a^{-1})(2 - a - a^{-1}) = a + a^{-1} - a^2 - a^{-2}.$$

All Fourier exponents in the last expression are mixed-sign, so q is orthogonal to $H^2(\mathbb{G}_2) + \overline{H^2(\mathbb{G}_2)}$, hence $q \in C(b\mathbb{G}_2) \cap W^\perp$.

Let $\mathcal{F}_N$ be product Fejer smoothing on $\mathbb{T}^2$. Since $\mathcal{F}_N(a) = \lambda_N a$ and $\mathcal{F}_N(a^{-1}) = \lambda_N a^{-1}$, where $\lambda_N = (1 - \frac{1}{N+1})^2 < 1$, the smoothed pullback is

$$\mathcal{F}_N(q \circ s) = 1 + \lambda_N(a + a^{-1}).$$

Multiplying again by the Szego weight gives a constant Fourier coefficient

$$\mathcal{F}_N(\widehat{q \circ s}|z_1 - z_2|^2)(0, 0) = 2(1 - \lambda_N) \neq 0.$$

Thus $\mathcal{F}_N(q \circ s)$ is not orthogonal even to the constant function in $H^2(\mathbb{G}_2)$. Ordinary torus Fejer smoothing therefore does not preserve the quotient determining annihilator.

Thus a full symmetrized-polydisc equivalence by this route reduces to a new quotient-boundary smoothing theorem: every positive determining measure for the Szego Hardy space of $\mathbb{G}_d$ should be a weak-star limit of positive continuous determining densities whose Cayley transforms can be rationalized on the quotient. That theorem is not present in the available source or quotient-domain literature checked here. This is the precise remaining barrier, not the atomic obstruction itself.

Consequence for the Source Problem

The result rules out the following direct implication strategy: starting from the Koranyi–Pukanszky Herglotz representation on $\mathbb{D}^d$, approximate an arbitrary determining probability measure by finite convex combinations of point masses that remain in the determining class, and then take inverse Cayley transforms to obtain rational inner approximants. Such finite determining point-mass combinations do not exist for $d \geq 2$. More strongly, the representing measures in the Koranyi–Pukanszky class are forced to be diffuse and to have Haar pushforwards along all mixed-sign characters.

This is only an obstruction to the atomic proof route, not a disproof of equivalence. The Clark-density reformulation above shows exactly what must replace atoms: rational-inner Clark measures, all of which are diffuse in the Koranyi–Pukanszky setting. A non-circular proof of the Herglotz-to-Caratheodory implication from the integral representation would amount to proving this Clark-measure density directly from the structure of $\text{RP}(\mathbb{T}^d)$. For the symmetrized polydisc, the analogous missing ingredient is the quotient-boundary smoothing theorem described above.

Verification and Novelty Notes

The proof uses only the Koranyi–Pukanszky mixed Fourier moment condition quoted in the source paper, the standard uniqueness theorem for finite Borel measures on $\mathbb{T}$, and the quotient-domain Caratheodory theorem of Bhowmik–Kumar [3]. A bounded search checked the source paper, the published arXiv/BLMS metadata, and web queries for the exact phrases “Herglotz representation”, “Caratheodory approximation”, “polydisc”, “symmetrized polydisc”, and “Clark measures”. The Clark-measure literature is consistent with the obstruction: multivariable Clark measures are positive RP-measures and are singular for inner functions [7], while bidisc rational-inner Clark measures are known not to have point masses [5]. The 2025 rational-approximation paper of Bhowmik–Putinar [6] gives a strong positive result for Agler-class Herglotz functions and, on the bidisc, for all Herglotz functions via Ando’s theorem. The BLMS page for the source paper also records a 2025 addendum clarifying that one theorem in the 2024 paper was previously known [2]. No later exact solution to the full polydisc/symmetrized-polydisc equivalence question was found in this bounded check. This packet should be reviewed as a strengthened partial obstruction and as a candidate full polydisc subcase, not as a full solution of the original two-domain question.

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